Chapter B, Problem 1PP

Problem

Convert the following complex numbers to polar and exponential forms:

(a) *z*1 = 3 − *j*4, (b) *z*2 = 5 + *j*12, (c) *z*3 = −3 − *j*9, (d) *z*4 = −7 + *j*.

Step-by-step solution

Step 1 of 5

Write the generalized polar form representation of complex number, $z = x + jy$.

$$z = r\angle\theta \quad \left(r = \sqrt{x^2 + y^2},\ \theta = \tan^{-1}\left(\frac{y}{x}\right)\right)$$

Note that here calculation of θ value changes depending on the quadrant of the given complex number.

Write the generalized exponential form representation of complex number, $z = x + jy$.

$$z = re^{j\theta} \quad \left(r = \sqrt{x^2 + y^2},\ \theta = \tan^{-1}\left(\frac{y}{x}\right)\right)$$

Note that here calculation of θ value changes depending on the quadrant of complex number.

Step 2 of 5

(a)

Consider the complex number, $z_1 = 3 - j4$.

Calculate the magnitude of complex number z_1.

$$\begin{aligned} r_1 &= \sqrt{3^2 + 4^2} \\ &= \sqrt{9 + 16} \\ &= \sqrt{25} \\ &= 5 \end{aligned}$$

Calculate the angle θ_1, as the complex number $z_1 = 3 - j4$ fall in fourth quadrant.

$$\begin{aligned} \theta_1 &= 360° - \tan^{-1}\left(\frac{4}{3}\right) \\ &= 360° - 53.15° \\ &= 306.84° \end{aligned}$$

Thus, the polar form of the complex number $z_1 = 3 - j4$ is $\boxed{5\angle 306.84°}$ and the exponential form of the complex number $z_1 = 3 - j4$ is $\boxed{5e^{j306.84°}}$.

Step 3 of 5

(b)

Consider the complex number, $z_2 = 5 + j12$.

Calculate the magnitude of complex number.

$$\begin{aligned} r_2 &= \sqrt{5^2 + 12^2} \\ &= \sqrt{25 + 144} \\ &= \sqrt{169} \\ &= 13 \end{aligned}$$

Calculate the angle θ_2, as the complex number $z_2 = 5 + j12$ fall in first quadrant.

$$\begin{aligned} \theta_2 &= \tan^{-1}\left(\frac{12}{5}\right) \\ &= 67.38° \\ &= 67.38° \end{aligned}$$

Hence the polar form of the complex number, $z_2 = 5 + j12$, is $\boxed{13\angle 67.38°}$ and the exponential form of the complex number, $z_2 = 5 + j12$, is $\boxed{13e^{j67.38°}}$.

Step 4 of 5

(c)

Consider the complex number, $z_3 = -3 - j9$.

Calculate the magnitude of complex number z_3.

$$\begin{aligned} r_3 &= \sqrt{3^2 + 9^2} \\ &= \sqrt{9 + 81} \\ &= \sqrt{90} \\ &= 9.487 \end{aligned}$$

Calculate the angle θ_3, as the complex number fall in third quadrant,

$z_3 = -3 - j9$ (3rd quadrant).

$$\begin{aligned} \theta_3 &= 180° + \tan^{-1}\left(\frac{9}{3}\right) \\ &= 180° + 71.56° \\ &= 251.56° \end{aligned}$$

Hence the polar form of the complex number, $z_3 = -3 - j9$, is $\boxed{9.487\angle 251.56°}$ and the exponential form of the complex number, $z_3 = -3 - j9$, is $\boxed{9.487e^{j251.56°}}$.

Step 5 of 5

(d)

Consider the complex number, $z_4 = -7 + j$.

Calculate the magnitude of complex number z_4.

$$\begin{aligned} r_4 &= \sqrt{7^2 + 1^2} \\ &= \sqrt{49 + 1} \\ &= \sqrt{50} \\ &= 7.071 \end{aligned}$$

Calculate the angle θ_4, as the complex number fall in second quadrant,

$z_4 = -7 + j \quad (\text{2nd quadrant})$.

$$\begin{aligned} \theta_4 &= 180° - \tan^{-1}\left(\frac{1}{7}\right) \\ &= 180° - 8.13° \\ &= 171.86° \end{aligned}$$

Hence the polar form of the complex number, $z_4 = -7 + j$, is $\boxed{7.071\angle 171.86°}$ and the exponential form of the complex number, $z_4 = -7 + j$, is $\boxed{7.071 e^{j171.86°}}$.